



matter

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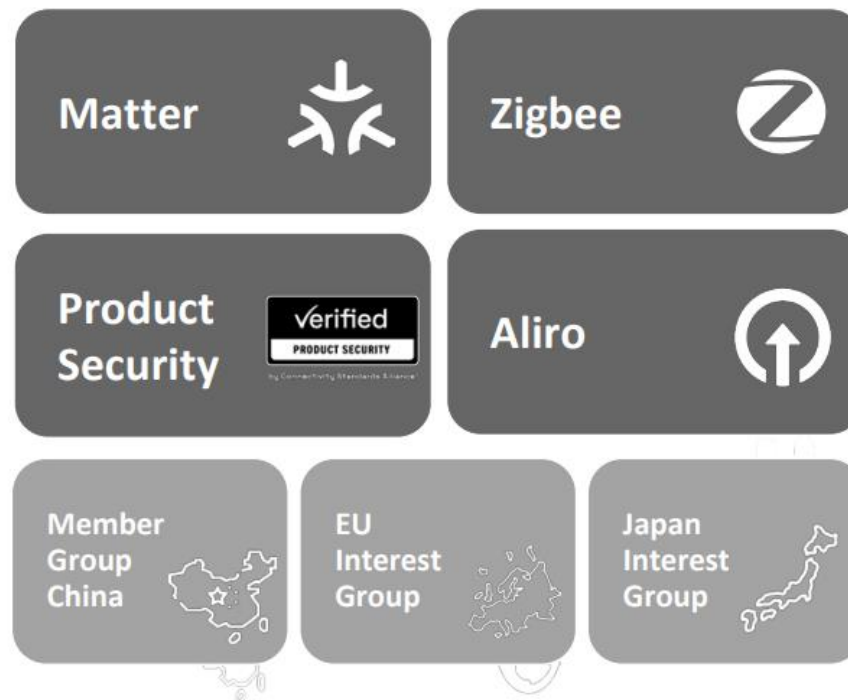
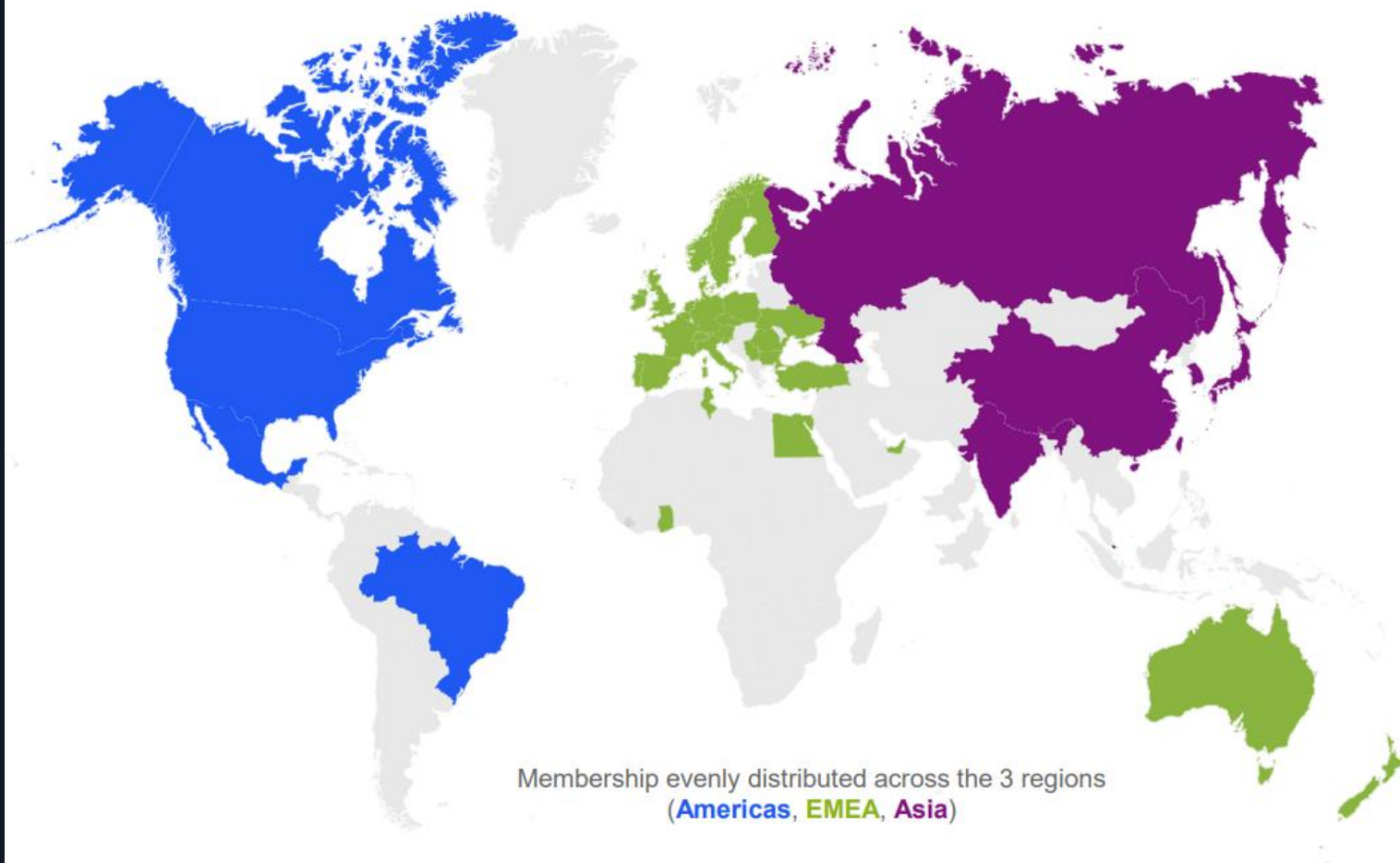
AGENDA

1. CSA organization
2. Network Topology
3. Network Stack
4. Device Data Model
5. Interaction Model
6. Commissioning

Connectivity Standards Alliance (CSA)

Our Global Community

700+ Member Companies and **9,000** individuals participating globally across **48** countries*



**As of November 2024*

Addressable market

The entire industry is invested.

The **300 members of the Matter Working Group** have brought their technology, best practices, business models — and commitment.

CNN

Smart homes see big breakthrough in CES 2022

WIRED

Here's how it could change the smart home scene.

The New York Times

Smart Home's Biggest Problem Might Get Better

TIME

The Best Inventions of 2021

THE VERGE

Matter's plan to save the smart home

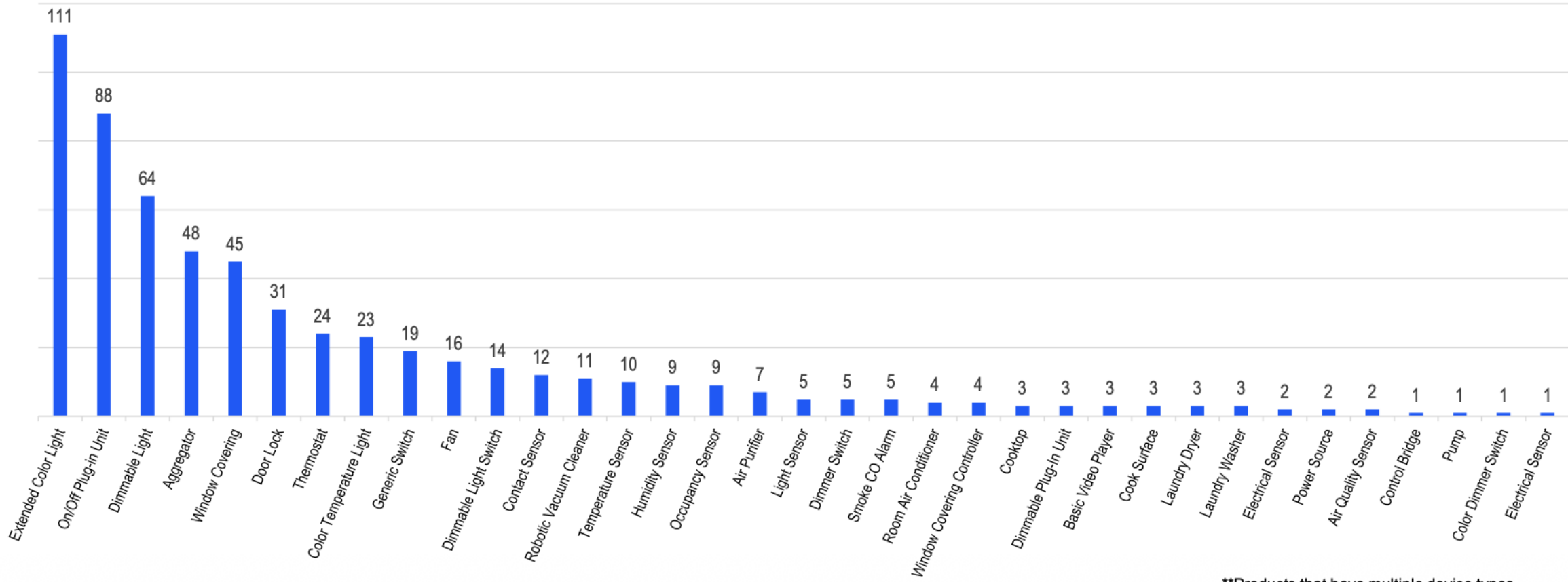




Matter End Product Device Types**

Last 12 Months (Through 31-January-2025)

*Unique Products Including CbS and Family



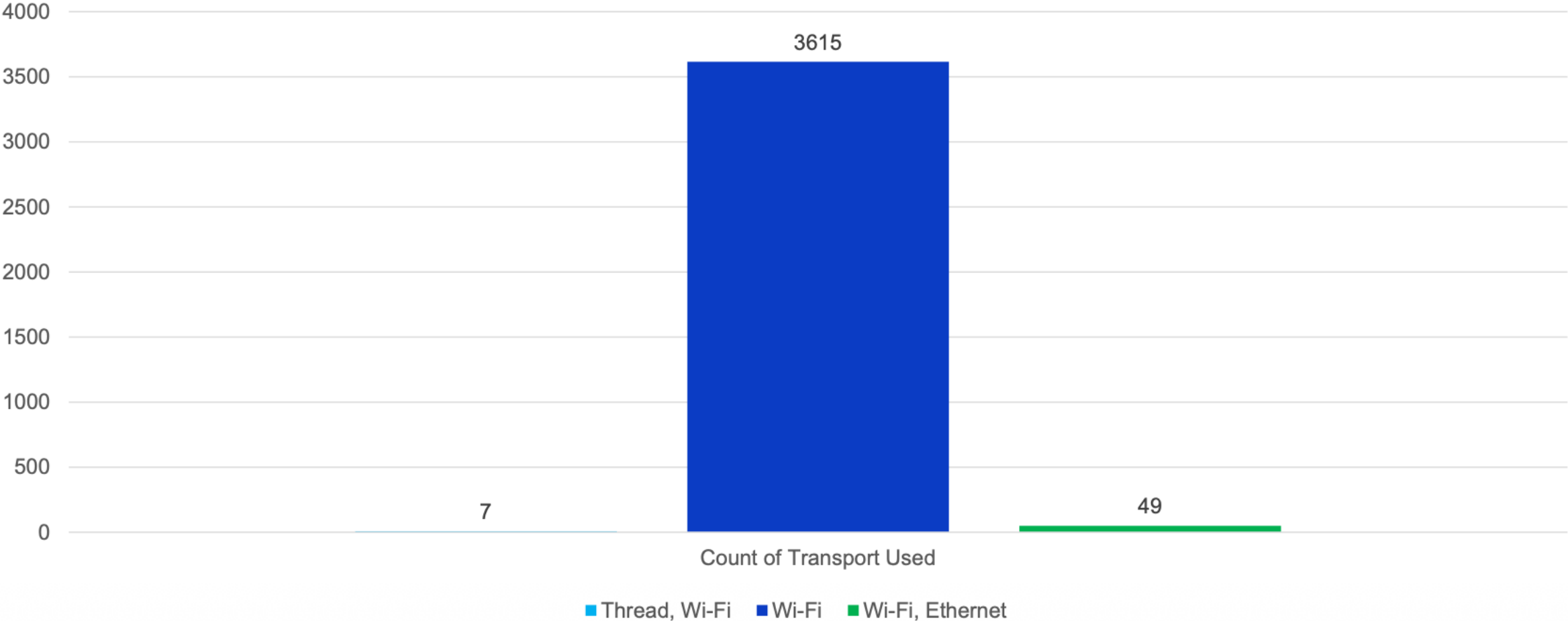
**Products that have multiple device types were counted for each type.





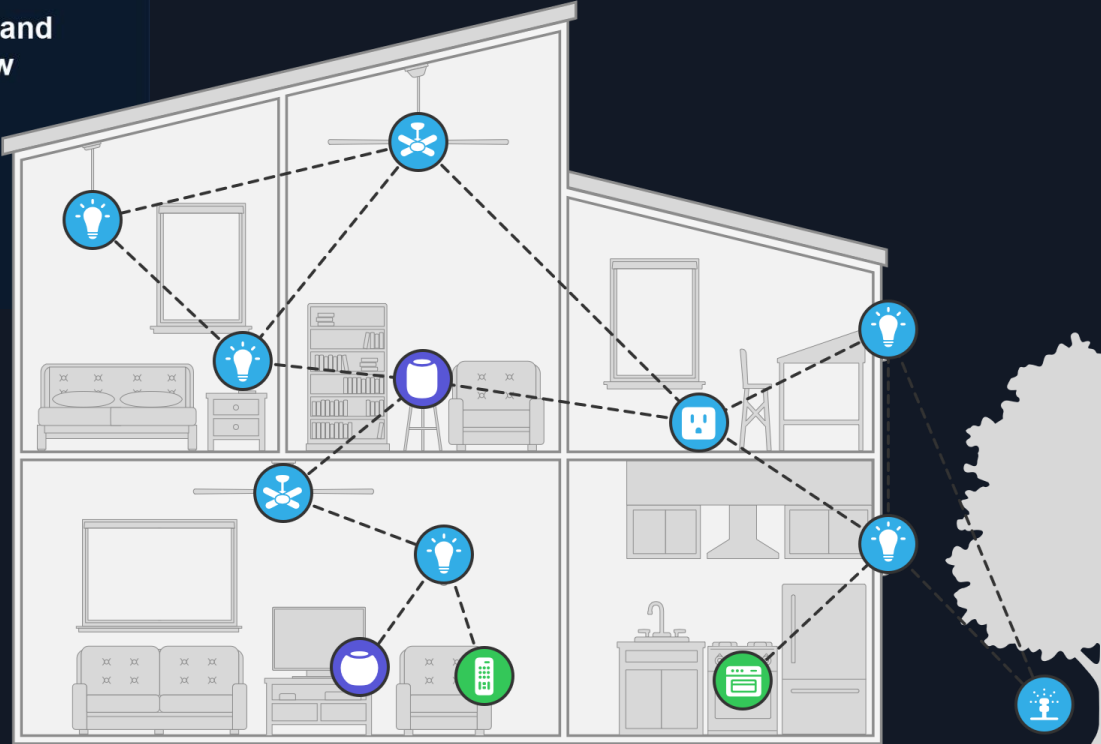
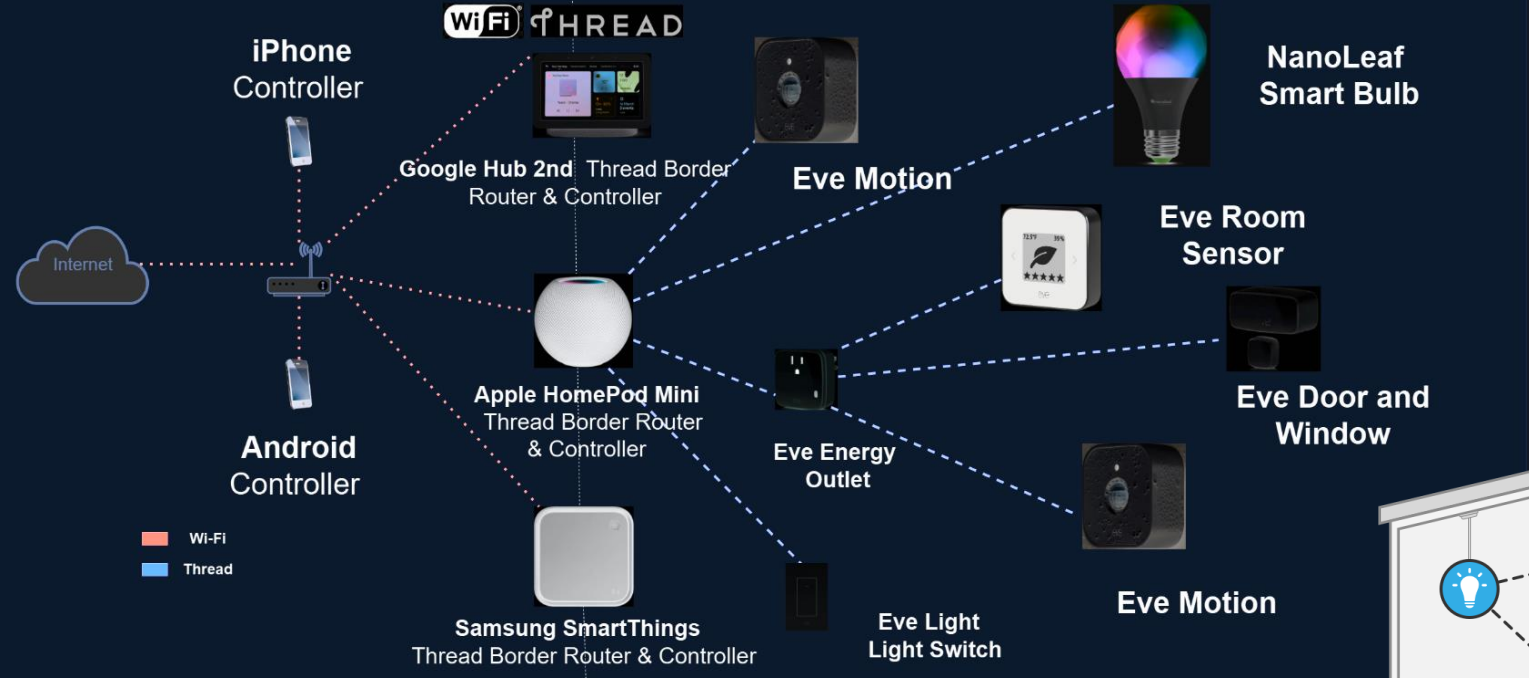
Matter Transports Lifetime

Certified Products with each Transport Type(s)



Network Topology

Topology Example



Network Topology

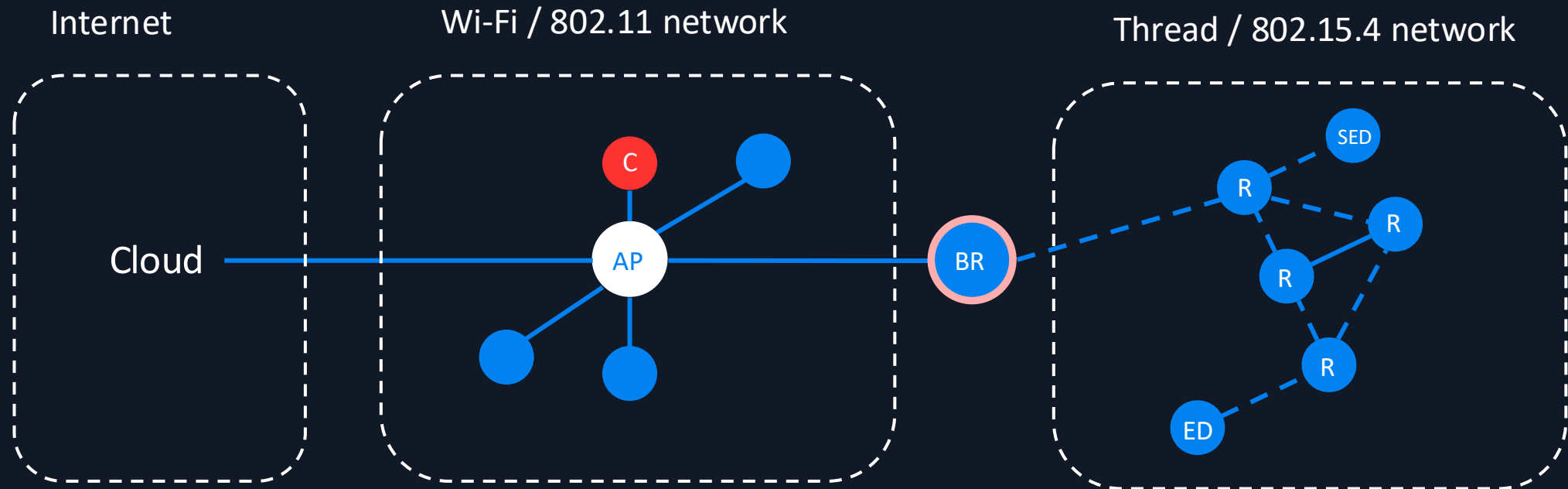
Fabric

- 64-bit fabric ID
- Root CA
- Multi-admin

Node

- 64-bit node ID
- NOC- Device ID

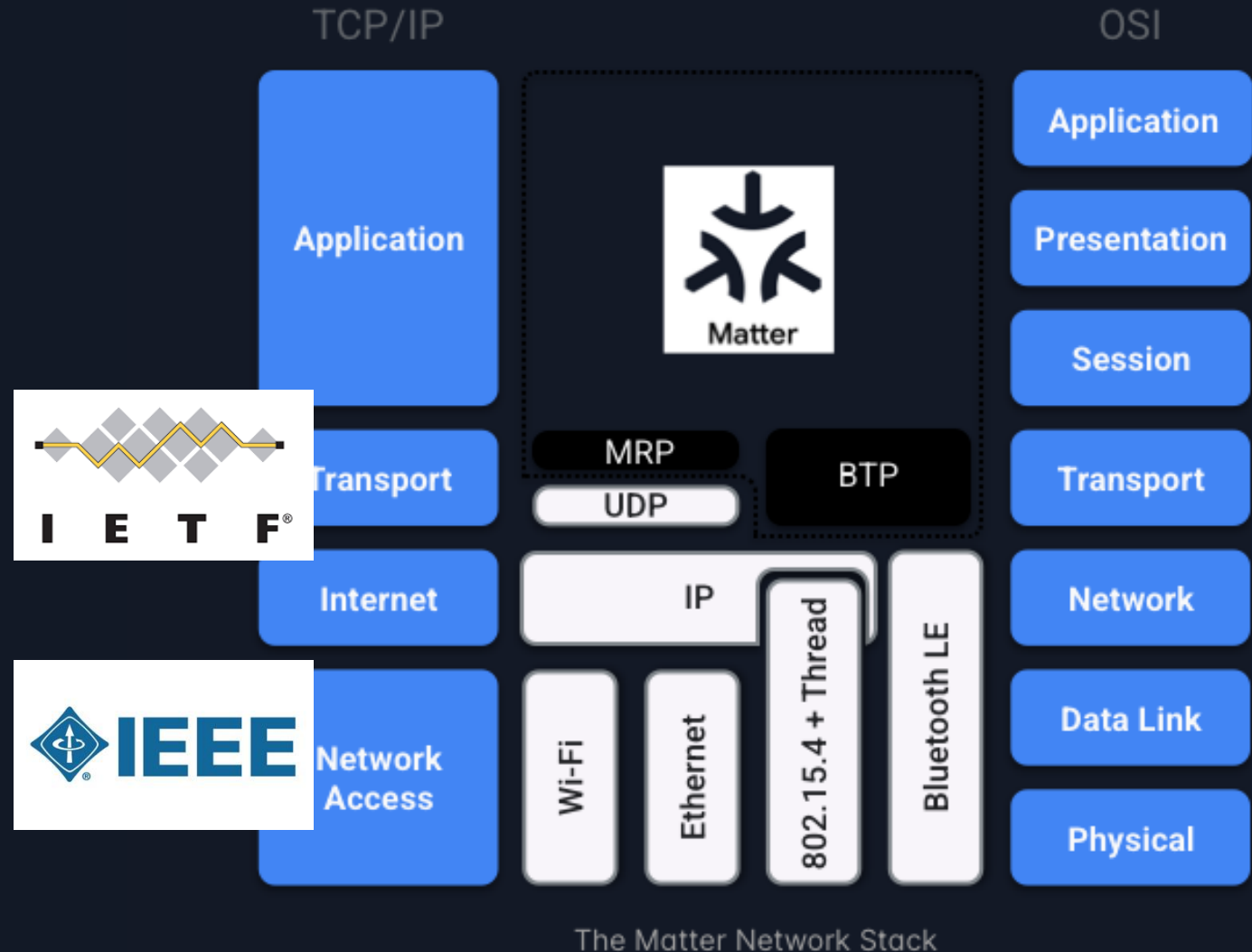
C	Commissioner	Mains - performs device onboarding
AP	Access Point / Hub	Mains - Wi-Fi infrastructure
BR	Border Router	Mains - bridges Thread and IP network
R	Router	Mains - extends Thread mesh
ED	End Device	Mains/Battery - leaf node
SED	Sleepy End Device	Battery - duty-cycled, wakes periodically



Network Stack

Overview: Layered stack

- Matter = Application Layer
- Transport-agnostic
- IPv6 only
- Wifi/Ethernet/Thread
- MRP layer for reliability over UDP
- 128 kB of RAM and 1 MB of Flash



Network Stack: Message framing

Every Matter Message contains:

- Message Header (unencrypted)

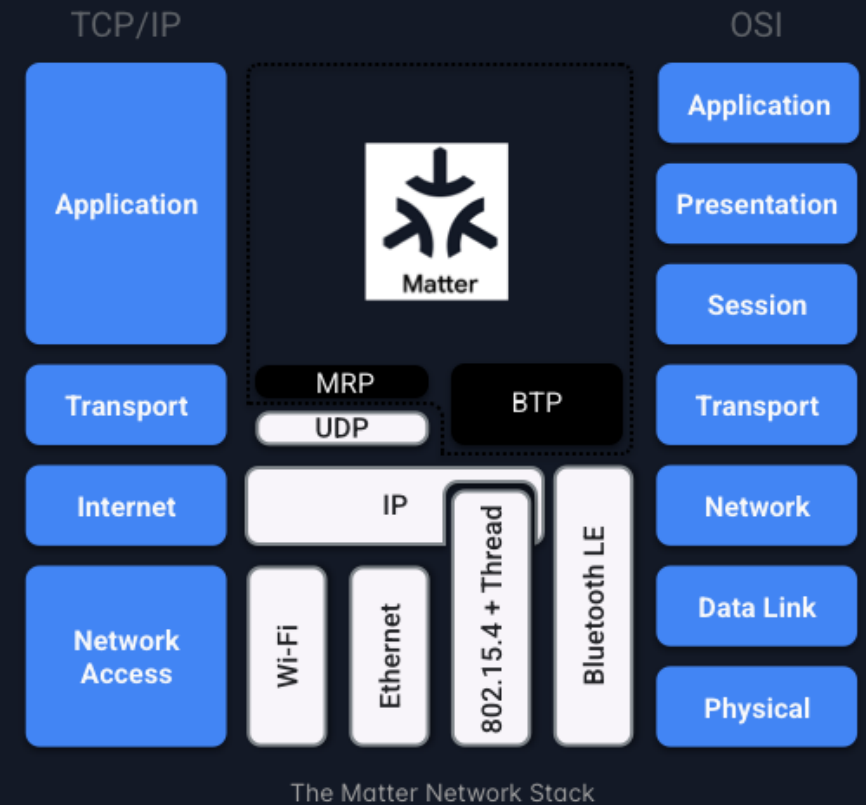
Message Flags (1 byte)	Session ID (2 bytes)	Security Flags (1 byte)	Msg Counter (4 bytes)	Node IDs (0-16 bytes)
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- Message Payload (encrypted)

Exchange Flags (1 byte)	Opcode (1 byte)	Exchange ID (2 bytes)	Protocol ID (2 bytes)	Application Payload (variable)
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- Message Footer

MIC - Message Integrity Check (variable, from AES-CCM)
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Network Stack: Transport & Discovery

Transport

- UDP port 5540 with MRP (Message Reliability Protocol) for ACKs/retransmissions
- TCP optional for large payloads
- BLE (BTP) for commissioning only

Addressing

- IPv6-only, no IPv4
- Link-local + ULA + routable addresses
- Multicast:

FF35:0040:FD<FabricID>00:<GroupID>

DNS-SD (RFC 6763) for all discovery.

- Commissionable node discovery: find new devices to commission
- Operational discovery: find commissioned nodes by Fabric ID + Node ID
- Commissioner discovery: commissionees find commissioners

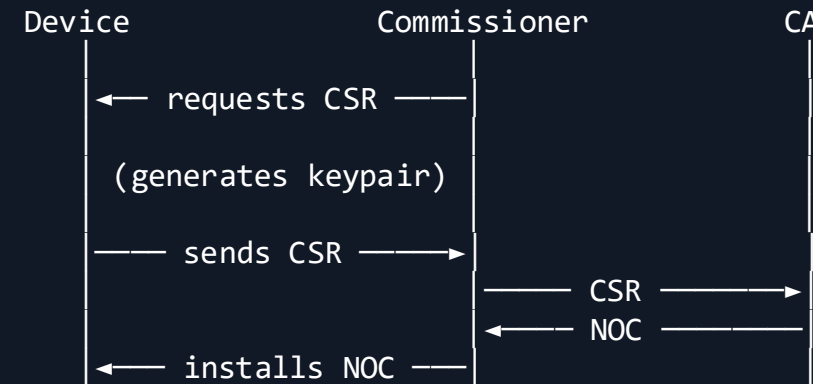
Thread-specific

- Unicast DNS via SRP, Border Router provides Advertising/Discovery Proxy

Network Stack: Security

- All Matter communication is encrypted and authenticated.
 - AES-128-CCM authenticated encryption
 - Message Integrity Code (MIC) appended to every frame
- No DTLS. Keys are derived from session establishment:
 - PASE (commissioning): Passcode-Authenticated Session
 - CASE (operational): Certificate-Authenticated Session

- Fabric trust model
 - Devices share a common Root of Trust per Fabric
 - Device generates its own key pair - private key never leaves device
 - NOC binds: Node ID + Fabric ID + public key
 - Multi-admin: one device can have NOCs from multiple Fabrics



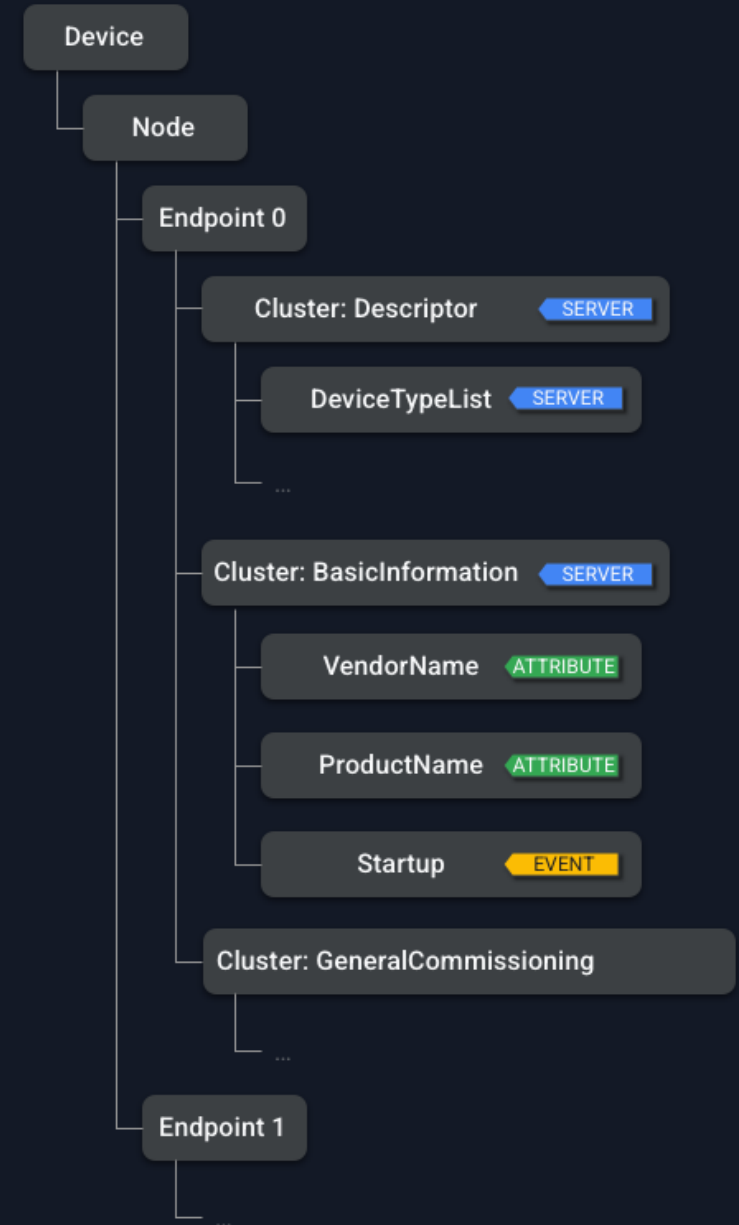
Device Data Model

Data Model

- Hierarchical modeling of a device's features.

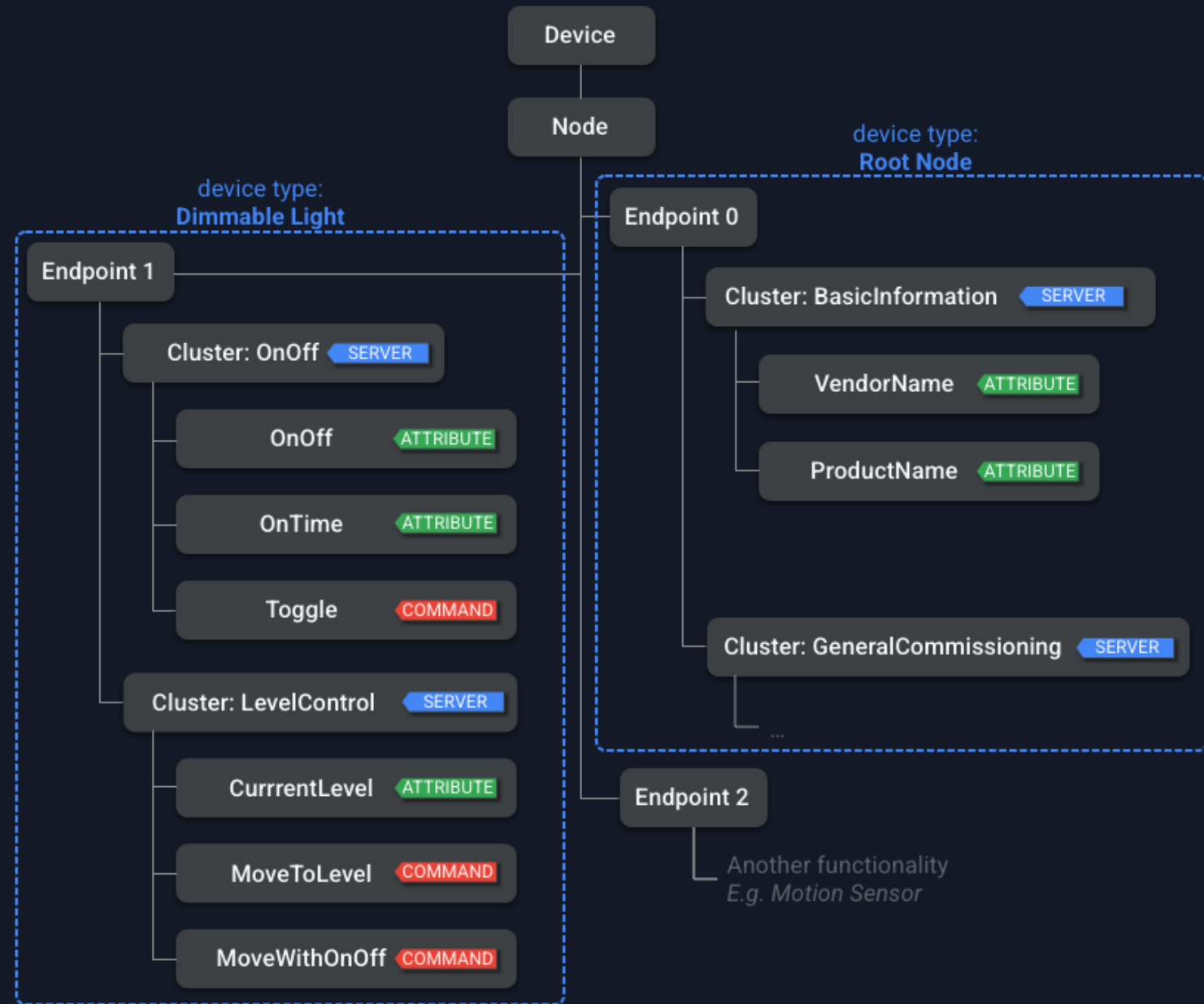
Device > Node > Endpoint > Cluster > Attributes/Commands/Events

- Endpoint 0 is always the root node endpoint - describes the device itself.
- Nodes have roles: **controller, controlee, OTA Provider, OTA Requestor**
- A node can have multiple roles
 - Smart speaker = Commissioner + Controller + OTA Provider
 - Light bulb = Controlee + OTA Requestor
 - Light switch = Controller only (cannot be controlled)



Clusters

- Grouping of related functionality within an Endpoint.
- Cluster Elements
 - Attributes: device state (nouns)
 - Commands: actions to perform (verbs)
 - Events: record of past state transitions (journal)



A sample of the hierarchy of Matter Devices interaction model

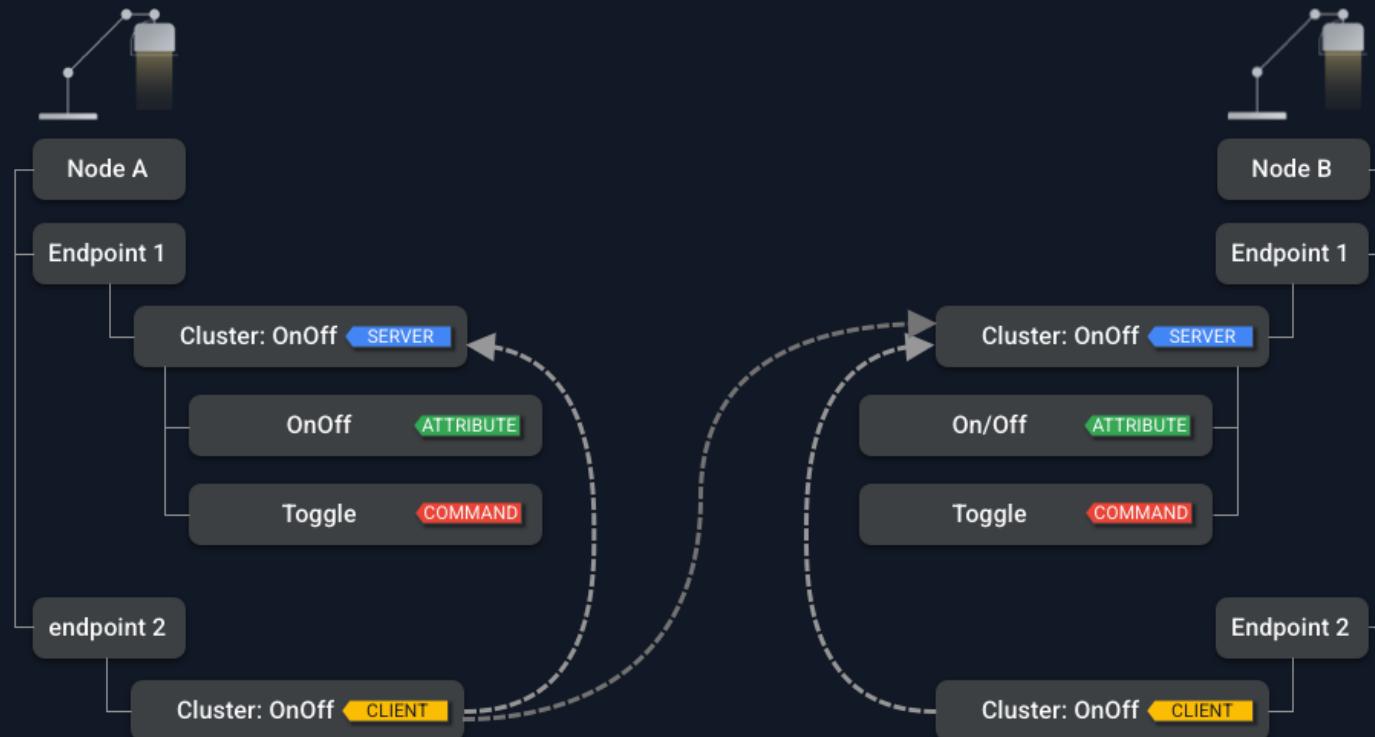
Client and Server Clusters



Server Cluster

Client Cluster

State	Stateful - holds Attributes, Events, Commands	Stateless
Role	Responds to requests	Initiates Interactions
Actions	Exposes data and accepts commands	Reads, writes, invokes on remote Servers



Interaction Model

Interaction Model: Concepts



Nodes interact with each other by...

- Reading and Subscribing to Attributes and Events (state monitoring)
- Writing to Attributes (configuration/settings)
- Invoking Commands (operational control)

Terms

Initiator	Client cluster, starts transaction
Target	Server cluster, responds
Group	16-bit ID, IPv6 multicast

Hierarchy

```
Interaction
├── Transaction(s) [1..n]
│   └── Action(s) [1..n]
```

Path assembly (no actual URLs)

```
<path> = <node> <endpoint> <cluster> <attribute | event | command>
<path> = <group ID> <cluster> <attribute | event | command>
```

Interaction Model: Concepts (path examples)

Path	Description
Node 0x1234 Endpoint 1 OnOff (0x0006) OnOff Attribute (0x00)	Read the on/off state of a light
Node 0x1234 Endpoint 1 OnOff (0x0006) Toggle Command (0x02)	Toggle a specific light
Node 0x1234 Endpoint 1 LevelControl (0x0008) MoveToLevel Command	Dim a bulb to a specific level
Node 0x1234 Endpoint 0 BasicInfo (0x0028) StartUp Event (0x00)	Subscribe to startup events
Group 0x0101 OnOff (0x0006) On Command (0x01)	Turn on all devices in a group
Group 0x0101 / LevelControl (0x0008) / MoveToLevel Command (0x00)	Dim all lights in a group

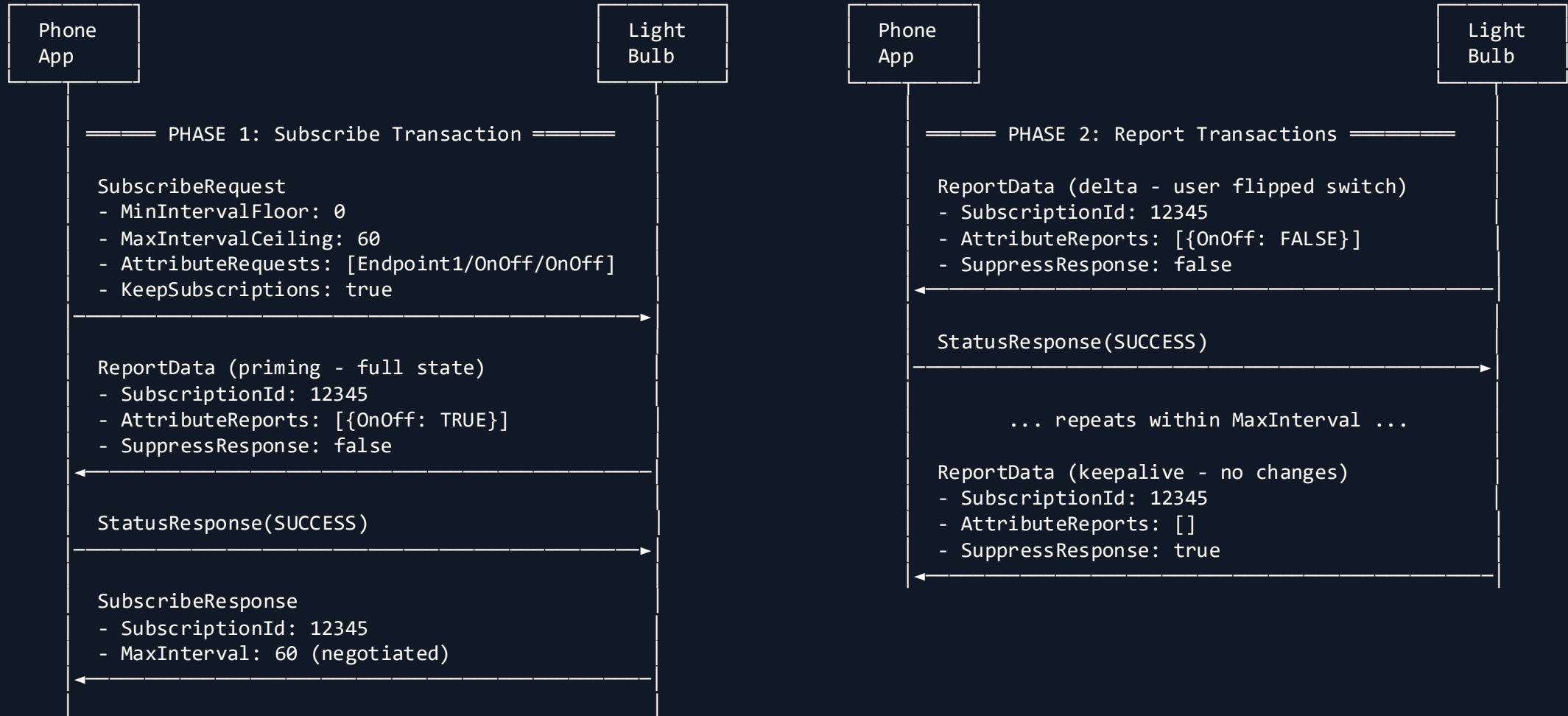
Interaction model: Read

- One shot request to get current state
- Status Response can be used for ACK on either direction (target-initiator, initiator-target).
- ReportData can Suppress Response
- Unicast-only - cannot target Groups
- Supports wildcards: Endpoint*/OnOff/OnOff reads all endpoints



Interaction model: Subscribe

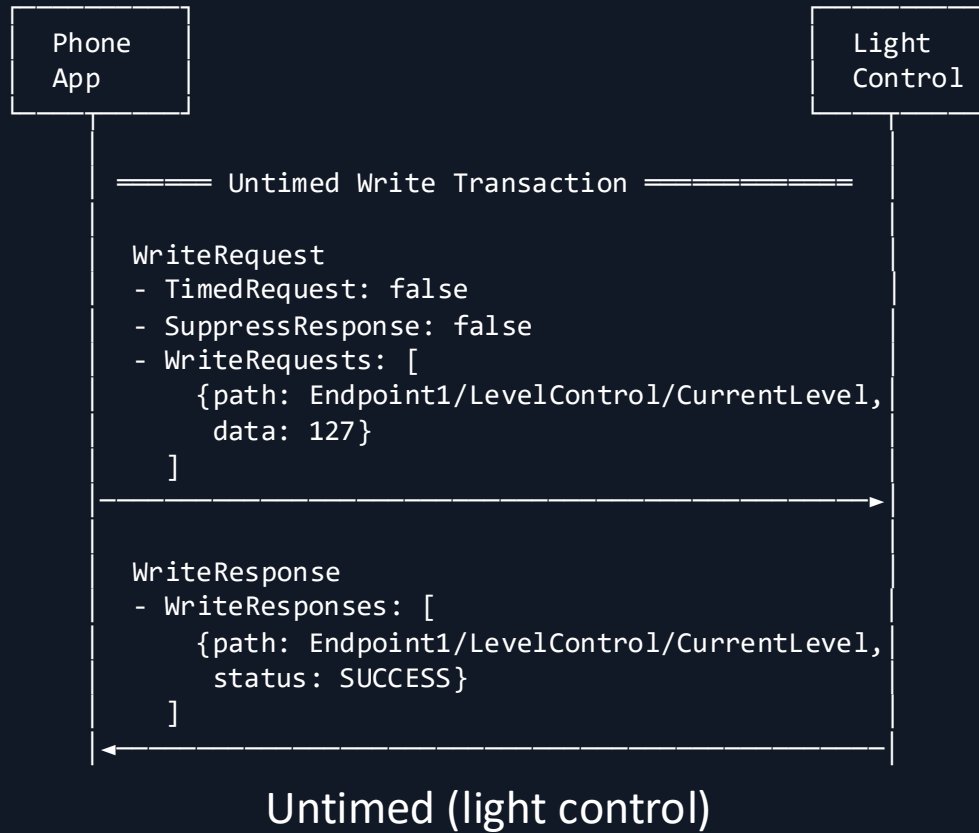
Persistent connection for ongoing updates.



Interaction model: Write

Modify attribute values.

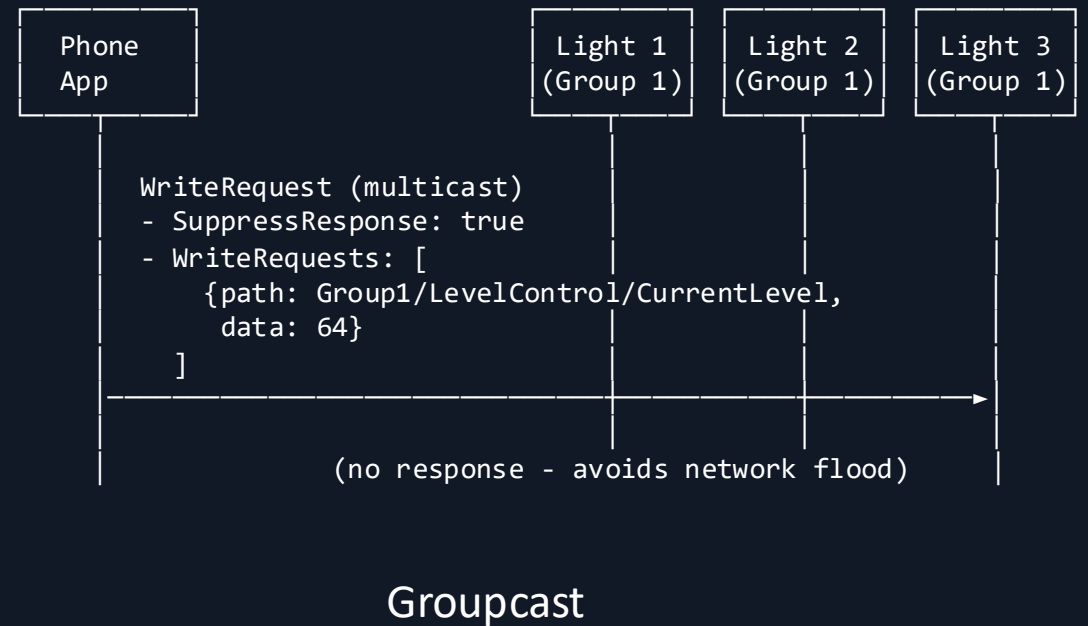
Two variants: timed and untimed (see next slide).



Groupcast requires SuppressResponse: true

WriteResponse returns status for each path in the request

Wildcards allowed on Endpoint field (e.g.,
Endpoint*/LevelControl/CurrentLevel)

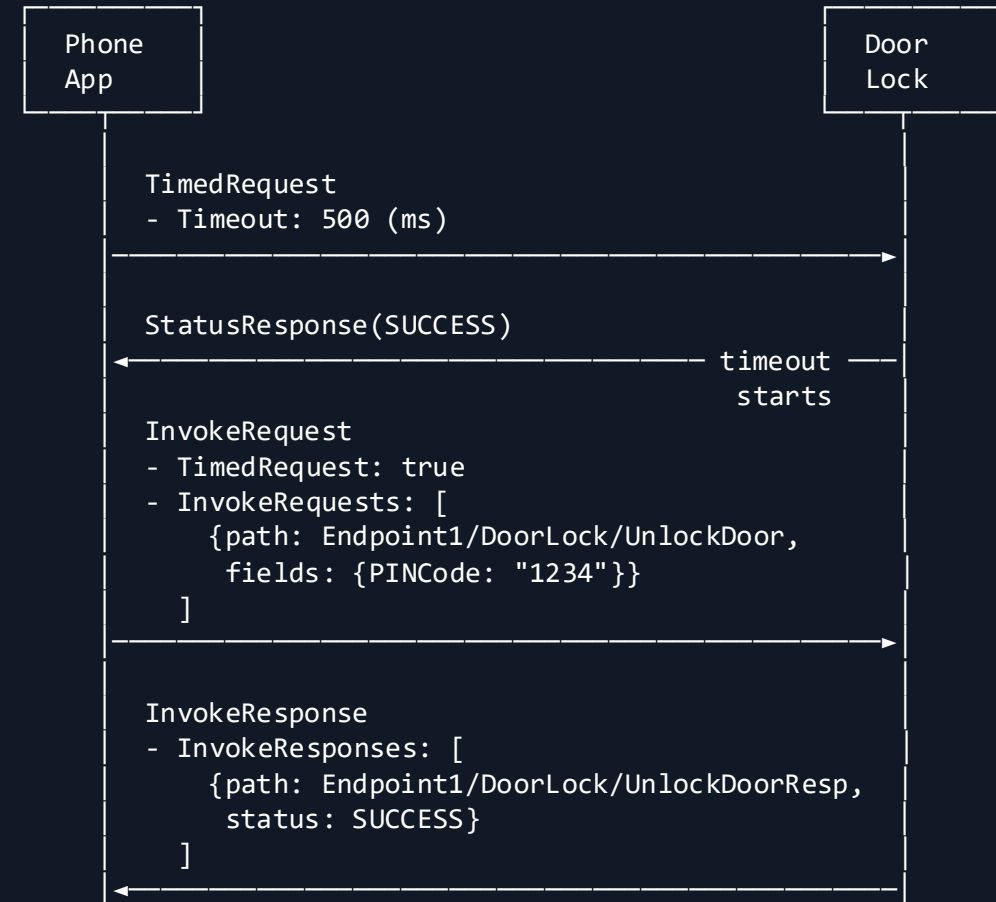


Interaction model: Invoke

Execute commands (cluster-defined actions with request/response semantics).



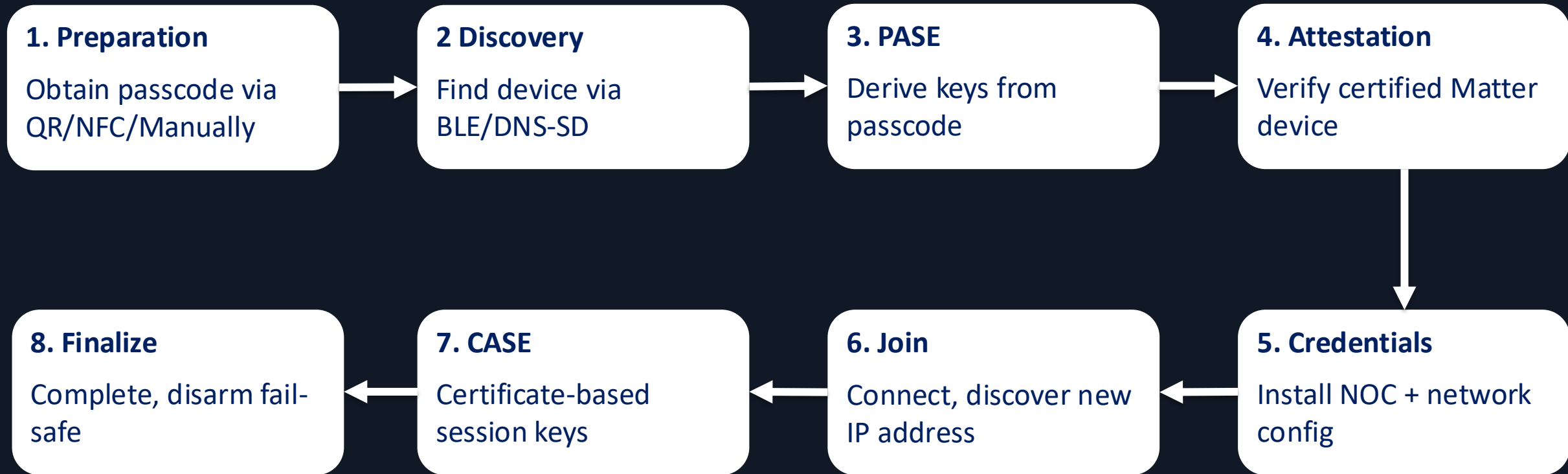
Untimed (light)



Timed (door lock)

Commissioning

Commissioning Flow





Handbook

<https://handbook.buildwithmatter.com/development/>

Google Matter Primer

<https://developers.home.google.com/matter/primer>

OpenThread Border Router Guide

<https://openthread.io/guides/border-router>

OpenThread Thread Primer

<https://openthread.io/guides/thread-primer>